

23. Ans: C

Magnitude of emf induced = rate of change of magnetic flux linkage

$$\begin{aligned}|E| &= \frac{\Delta\Phi}{\Delta t} \\&= \frac{NA\Delta B}{\Delta t} \\&= \frac{N\pi\left(\frac{d}{2}\right)^2 \Delta B}{\Delta t} \\&= \frac{(120)\pi\left(\frac{0.30}{2}\right)^2 (0.080 - 0.020)}{4.0} \\&= 0.127 \text{ V} \\&\approx 130 \text{ mV}\end{aligned}$$

(Even when the ends of the coil are not connected to form a closed circuit, there is still an emf induced in it due to the decreasing magnetic flux linkage.)

24. Ans: B